

Dictionary Learning as Unsupervised Steering-Vector Generation: Ontologizers vs. Sparse Autoencoders on SONAR Sentence Embeddings

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Abstract

The Ontologizer is a multihead, multilayer dictionary-learning architecture whose design goal is *unsupervised steering-vector generation*: a stack of softmax-competitive classifier “heads” reconstructing an embedding space should discover the space’s natural contrastive clusters, MoEfyng the representation into small, nameable classification tasks whose within-head entry swaps are steering moves. We evaluate this against field-standard top- k sparse autoencoders (SAEs) trained on identical cached SONAR multilingual sentence embeddings under an identical whitened reconstruction objective, using a common battery: a reconstruction-vs-capacity Pareto, categorical-form and semantic-coherence tests of head structure, cross-model feature splitting and seed stability, downstream text fidelity through the paired M2M100 decoder, sparse language probing, shrinkage decomposition, steering fidelity, and blinded auto-interpretability scoring. A *structural-ablation ladder* — SAE variants adding one Ontologizer-style commitment at a time — attributes effects to specific structure. Headline preliminary findings: (i) contrast-set structure is expensive in reconstruction — the plain-SAE frontier dominates the Ontologizer’s deviation code at every capacity, and the grouped-simplex constraint alone costs $\sim 30\times$ in FVU_w ; (ii) trained heads carry the signature the steering goal predicts — within-head decode directions *repel* (96% of Ontologizer heads below a random-group null), i.e. heads form well-separated contrast sets rather than semantic clusters, in both the Ontologizer and its single-layer grouped-softmax analogue; (iii) categorical (exhaustive/exclusive) form exists precisely when trained in as hard selection and never emerges spontaneously or post hoc; (iv) per-latent evaluation lenses — k -sparse probing, per-feature monosemanticity — systematically mis-measure classifier codes, whose content lives in partitions rather than latent magnitudes; this extends to blinded detection, where per-latent describability tracks code density rather than architecture (the near-dense control matches the Ontologizer) and parameter-space decodes of feature directions fail as descriptions outright; and (v) a single-layer grouped-softmax SAE reproduces the Ontologizer’s code structure, contrastive signature, and text fidelity at better reconstruction with half the parameters — but it is the *worst* steerer in the battery, essentially never producing a strong intervention effect, so the full architecture wins the one benchmark the steering thesis is staked on; task naturalness remains unrun. Steering also dissolves the trade-off it was built to measure: collateral is direction-independent at matched magnitude, so there is no effect-vs-collateral frontier, and codes separate on effect at *low* magnitude — where hard grouped selection beats every soft code, including the Ontologizer.

1 Evaluation frame

The comparisons below were largely built on SAE-native terms — per-feature monosemanticity, sparse probing, reconstruction at matched sparsity — and we report them unchanged. But the Ontologizer’s goal is not a monosemantic dictionary. It is an *unsupervised generator of steering vectors*: each head is a k -way classifier over its input, trained only by reconstruction pressure, and the hope is that a series of such classifiers finds the most natural contrastive clusters in the embedding space. Interpretability value is then expected from *MoEfication*: decomposing the space into small classification tasks — one per head — each nameable in isolation, with within-head entry swaps as the steering primitive (forcing head h from its argmax entry to entry k' moves the decode along exactly the dimension head h classifies).

Under this frame the evaluation questions are: (a) what does contrast-set structure cost in reconstruction (the Pareto and ladder); (b) do trained heads actually behave as contrastive classifiers (categorical form, semantic anti-coherence); (c) are the learned tasks *natural* — nameable, stable across runs, aligned with generative factors of the data (task reconstruction, partition-level stability; largely future work); and (d) is the steering primitive good — consistent effect, low collateral, at least matching supervised diff-of-means vectors and SAE decoder-row addition (steering fidelity; preliminary). Metrics that ask whether *individual latents* are concept-aligned remain informative about the SAE side but are the wrong lens for classifier codes, and we flag them as such where they appear.

2 Related work

No single line of prior work covers this combination; each commitment has a nearest neighbour.

Discrete and routed dictionaries. A DictBlock head — a softmax over k entries, summed across heads — is soft product quantization, and the residual layer stack is soft residual VQ, the codec lineage of SoundStream [8]; that literature treats codes as compression and does not read codebooks as features. Codebook Features [9] is the closest interpretability ancestor (VQ bottlenecks finetuned into a network, hidden state a sum of a few discrete codes), but its codes are read per-code, not as classifiers, and there is no intervention grammar over within-codebook swaps. Switch SAEs [10] route activations among expert SAEs purely as a compute optimization; notably their experts collapse into redundant ensembles that duplicate features — exactly the failure mode `headstruct.py/headcoh.py` instrument, and the opposite of what trained heads show here (within-head directions repel). MoEfication [11] supplies the frame’s name but decomposes already-trained MLPs post hoc; here the partition is the training objective’s own discovery.

Structured SAEs. The field’s main structure axis is nesting by scale: Matryoshka SAEs [4] train nested dictionaries to curb feature absorption [5], and Tree SAE [12] learns explicit hierarchies. Rung 3 (`k32_p5`) imports exactly the matryoshka commitment into the ladder; the Ontologizer’s axis is instead *partition into exhaustive, mutually exclusive heads*. Both lines ask whether an architectural commitment buys interpretability that code density alone does not explain — the auto-interp results below answer “not per-latent describability,” while the description-coherence view is where the partition begins to pay.

Embedding-space substrate. SAEs over dense text embeddings became a subfield in 2025–2026: discretizing retrieval embeddings [13], data-analysis toolkits [14], human-concept alignment [15]. All are single-layer top- k SAEs, and none address SONAR, where the stakes are higher — Large Concept Models generate directly in SONAR space [16] — and where the paired M2M100 decoder supplies a semantic oracle: where embedding-inversion work must train an inverter [17], `textfid.py` and `steerfid.py` decode reconstructions and interventions through the pretrained decoder itself.

Evaluation stance. SAEBench [6] is the template for the multi-instrument battery (`langprobe.py` is deliberately SAEBench-style), and the blinded detection scoring follows the scaled auto-interp line [18]. AxBench [19] found that prompting and finetuning beat every representation-based steering method, with SAEs uncompetitive even at concept detection — independent support for judging dictionaries by steering utility rather than per-latent monosemanticity, and the niche `steerfid.py`’s effect-vs-collateral-at-matched-magnitude design targets. Song et al. [20] argue seed-to-seed feature consistency should be a first-class SAE metric; `splitting.py`’s match fractions measure precisely that. Without close analog, to our knowledge: reading grouped routing as contrastive classification with a causal intervention interface; quota-free coherence tests against size-matched nulls, including over judge-written description texts; and decoding parameter directions through the paired decoder as a test of whether features can name themselves from weights alone — a test the weights fail.

3 Setup

Embedding space and data. All models are trained on a fixed cache of $N=4,000,000$ SONAR [1] sentence embeddings ($d=1024$, unit-norm) computed from an interleaved 86-language mC4 sample, with a language label sidecar. The stream order is deterministic, so raw text is recoverable by row index (verified against the language sidecar per row). Because SONAR embeddings pair with an M2M100 text decoder, reconstructions and interventions can be decoded to text — steering outcomes are directly observable.

Objective and evaluation. All models minimize per-dimension inverse-variance-weighted (“whitened”) MSE with mean-one weights. We report FVU_w : whitened MSE divided by the whitened variance of a constant predictor, on the final 32,768 cache rows. This tail is excluded from SAE training. *Caveat: the Ontologizer trained on the full cache including the tail; a fresh-encoded evaluation set is future work.* Where activations are needed, Ontologizer features are the flattened per-head classification probabilities $P \in \Delta^{k \cdot l \cdot h}$ at temperature 0.03 (feature $f = (l \cdot h + h_i)k + k_i$; fire threshold $2/k$), and SAE features are latent activations (fire threshold 0; grouped-softmax runs use the above-uniform threshold $2/(m/G)$, matching the tag convention).

Models. The reference Ontologizer (`resid_nc`, step 372,600) has $l=5$ residual layers, $h=32$ heads, $k=32$ entries per head, bilinear ($n=2$) classifiers with residual normalization and a constant input coordinate, deep supervision, and a shared linear decoder from a 2048-d dictionary space; ≈ 23.1 M parameters and $l \cdot h \cdot k = 5,120$ dictionary entries. Layers 0–3 trained healthily (prefix FVU_w $0.129 \rightarrow 0.0051$); layer 4 froze 22/32 heads late in temperature annealing and is mildly harmful. SAEs follow Gao et al. [2]: pre-activation $\text{relu}((x - b_{dec})W_{enc} + b_{enc})$ with top- k selection, unit-norm decoder rows, tied initialization, aux- k dead-latent revival, trained

with Adam at 4×10^{-4} for $\sim 144\text{k}$ steps at batch 4096 (converged; convergence is optimizer-step-limited, not epoch-limited, at this batch size).

The structural-ablation ladder. To attribute differences to specific architectural commitments rather than to whole architectures, we train SAE variants that each add one Ontologizer-style constraint (all $m=5120$, same data and objective):

rung	run	added structure	analogue
1	k32	none (top- $k=32$)	baseline
2a	g160top1	160 groups, hard winner per group	heads (hard)
2b	g160softmax	160 groups, softmax per group	heads (soft)
3	k32_p5	5 nested prefix losses (matryoshka)	deepsup layers
—	k32_b1	bilinear encoder, $[x-b_{dec}; 1]$ factors	bilinear classifier

Controls at width $m=11,264$ (parameter-matched to the Ontologizer): **k32**, **k160** (coefficient-matched to one deviation per head), and **k5120** (coefficient-matched to the full soft code; the top- k constraint never binds — measured $L_0 \approx 1520$ — so it is effectively a dense ReLU autoencoder). A seed replica **k32_s43** supports stability analysis. The bilinear encoder’s latents admit closed-form “eigenfeature” analysis: each latent’s symmetric form $\text{sym}(w_1 w_2^\top)$ is rank 2, with top- $|\lambda|$ eigenvector $w_1/\|w_1\| \pm w_2/\|w_2\|$ (Pearce et al. [3]). Rung 2b (**g160softmax**) is the load-bearing control under the steering frame: it isolates the “series of softmax classifiers” idea from the residual stack, bilinear classifiers, and shared dictionary.

4 Methods

Each protocol is a standalone script sharing the eval split and activation conventions; all are unit-tested against planted-structure synthetic data.

Capacity Pareto (pareto.py). Capacity is continuous coefficients transmitted per sample. Ontologizer code points are *top- m deviation truncations*: per head, keep the m entries with largest $|p - \text{origin}|$, pin the rest to the origin distribution, renormalize. The origin is the corpus-mean classification $\mathbb{E}[p]$ — measured, not uniform: the model output is jointly *linear* in the flattened classifications (zero residual seed, linear decoder), so $\text{decode}(\mathbb{E}[p]) = \mathbb{E}[Y]$, which decodes to the corpus-mean generic text, while the uniform code lands off-manifold (cosine 0.067 to $\mathbb{E}[X]$). Endpoints: $m=0$ (input-independent origin), $m=k$ (exact soft model), and end-to-end argmax (“hard”). An alternative head-level code mean-ablates all but the m lowest-entropy heads per layer. SAE points use measured eval L_0 .

Categorical form (headstruct.py). An Ontologizer head is exhaustive (always fires) and exclusive (one winner). We test whether SAE latents form such units: discovery clusters latents by *paradigmatic* affinity — similar co-fire profiles with the rest of the dictionary $\times$ anti-correlation with each other ($\text{PMI} < 1$) — under a group-size cap, on one half of the rows; scoring on the held-out half computes per-group exhaustiveness $P(\text{any fires})$, exclusivity $\mathbb{E}[\#\text{firing} \geq 1]$, and the CV of summed activation, against size-matched random partitions of the live-latent pool.

Head-like structure requires exhaustiveness *above* null with exclusivity *below* null; topical co-firing clusters show the opposite signature. `--trained-groups` scores a grouped run’s own partition.

Semantic coherence vs. contrast (`headcoh.py`). The complement to categorical form: do a head’s entries *mean* similar things, or do they form a contrast set? Every entry has an exact decode direction in SONAR output space — for the Ontologizer, the rows of the linear map G with $Y = P_{\text{flat}}G$; for SAEs, decoder rows — and SONAR space is a semantic sentence space, so within-head direction similarity is semantic similarity of what entries decode to. Per head we report mean pairwise cosine and top-singular-value energy fraction, z -scored against size-matched random subsets of the same pool (same layer for Ontologizer heads); as a metric-hygiene control the same statistics are recomputed under the training objective’s inverse-variance inner product (`--metric whitened`). $z \gg 0$ means a head clusters semantically; $z \ll 0$ means its entries repel — the signature of a *contrastive classifier*, whose class decoders are pushed apart. A second view SONAR-encodes auto-interp description texts and tests pooled within-head similarity against a label-permutation null. `--assignment` scores discovered (post-hoc) groups.

Feature splitting and seed stability (`splitting.py`). For models A (coarse) and B (fine) on shared rows, each B -latent is classified per A -parent by two-way firing containment at threshold τ : a *child* satisfies $P(A_i | B_j) \geq \tau$, $P(B_j | A_i) < \tau$; a *match* satisfies both. Splits (≥ 2 children) are scored by union coverage of the parent’s firing set and pairwise child overlap. Decoder geometry adds absorption candidates (decoder-cosine twins that do not co-fire) and each model’s nearest-neighbour decoder-cosine distribution vs. a random baseline. For two same-architecture runs differing only in seed, the match fractions are the reproducible-feature fractions.

Text fidelity (`textfid.py`). The “loss recovered” analogue: decode x and $\text{recon}(x)$ through M2M100 (forced `eng_Latn`, both rescaled to the reference SONAR norm, so the metric sees direction error) and report chrF [7] and exact-match between the decodes, plus the decoder’s teacher-forced per-token NLL of the *original’s* decode under the original (ceiling), the reconstruction, and the corpus-mean embedding (floor — which decodes to the generic text). Under the steering frame this doubles as the collateral ceiling: a code that cannot preserve text cannot steer text with low collateral.

Language probing (`langprobe.py`). SAEBench-style k -sparse probing with the free 86-language labels: top latents per language by two-sample t -statistic on training rows; held-out one-vs-rest F1 at $k=1$ (single latent, fitted threshold), $k=4, 16$ (logistic probes), and a dense probe on the raw embedding as ceiling. Macro-F1 over languages. *Frame caveat: this measures whether a factor lives in individual latent magnitudes. A classifier head carries its factor in its assignment (partition), which single-latent probes cannot see; partition-level mutual information is the matching instrument for grouped codes.*

Shrinkage (`refit.py`). Gao-style refinement: freeze each sample’s support, solve the whitened least-squares coefficients (masked normal equations; silent group slots forced to zero), and compare FVU_w . Raw-refit is magnitude (shrinkage) error; refit is selection error. The Ontologizer

shares the machinery via its exact linearity $Y = P_{\text{flat}}G$, with G measured by decoding one-hot constant codes; its supports are the top- m deviation entries. Points with support $\geq d$ are excluded (underdetermined). Note the raw convention: `refit.py` truncates the soft forward’s P (non-adaptive), whereas `pareto.py` truncates inside the forward, letting later layers reclassify against the truncated prefix residual.

Steering fidelity (`steerfid.py`). Unit-normalized native steering directions — SAE decoder rows, bilinear eigenfeatures, or the Ontologizer’s `withArgs` set-intervention delta — applied at matched embedding magnitudes. Effect is cycle consistency: the steered decode is re-encoded through SONAR and the target feature’s activation gain is normalized by its firing-conditioned quantiles, with the baseline taken from the *cycled* base decode so both sides share roundtrip degradation. Collateral is $1 - \text{chrF}$ against the base decode. Random unit directions give the control curve. Under the steering frame this is the primary benchmark; only smoke-scale results exist so far.

Intervention composition (`compose.py`). Treating heads as candidate causal variables: for pairs of forced assignments $(l_1, h_1) \rightarrow k_1$, $(l_2, h_2) \rightarrow k_2$ (8 random heads per layer, one random entry each; all same-layer pairs plus 60 cross-layer pairs), compare the joint intervention’s decode delta D_{12} against the sum $D_1 + D_2$ of the single-intervention deltas on 512 held-out rows: median per-row cosine and relative error $\|D_{12} - (D_1 + D_2)\| / \|D_1 + D_2\|$, raw and whitened. Because the decode is exactly linear in the flattened classification, final-layer pairs compose exactly by construction; that stratum doubles as the measured numerical floor (the $\|R\| \approx 31 \rightarrow \|Y\| \approx 1$ decoder cancellation amplifies float32 rounding to ~ 0.02 relative error between differently-compiled variants — verified exact in float64). All other non-additivity is downstream reclassification: later layers re-reading the intervened residual.

Auto-interpretability (`autointerp.py`). Four stages: harvest (top-activating rows, negatives, frequency-stratified feature selection, parameter-based decode embeddings), texts (deterministic mC4 stream replay recovers text by row index, verified against the language sidecar), describe (mode `params`: M2M100 decode of the parameter-derived embedding — for tags, the $\mathbb{E}[p]$ -origin code with one head forced one-hot; for SAE latents, $b_{\text{dec}} + \alpha W_{\text{dec},j}$; mode `eig`: the bilinear eigenfeature decode; mode `acts`: an LLM summary of top-activating snippets; mode `cacts`: a *contrastive* summary — the judge sees top snippets next to random corpus draws and must name what distinguishes them, blocking the degenerate corpus-level answer that plain summaries invite on noisy multilingual web text), and score (blinded detection over held-out positives and negatives, identical for all modes; per-feature F1, to be read against the always-match null of $F1 \approx 0.67$ on balanced items). The judge is Haiku (`claude-haiku-4-5`) via the Anthropic Messages API, rate-metered, with bounded reply length; failed judgements are never recorded (an earlier outage mode scored empty replies as $F1 = 0$). Six models: the Ontologizer, rungs `k32/k32_b1/g160top1/g160softmax`, and the dense control `k5120`.

point	coeffs	index bits	FVU _w
onto $m=0$ ($\mathbb{E}[p]$ origin)	0	0	1.001
onto hard (argmax)	0	800	17.34
sae k32 tier ($m=5120/11264$, incl. p5, s43)	32	~ 400	0.494–0.509
sae k32 bilinear	32	394	0.602
onto dev $m=1$	160	800	0.677
sae g160top1	160	1972	0.274
sae k160 ($m=11264$)	160	2154	0.258
onto dev $m=2$	320	1600	0.497
onto dev $m=4$	640	3200	0.289
onto dev $m=8$	1280	6400	0.094
sae k5120 ($m=11264$, $L_0 \approx 1520$)	1520	20455	0.00011
onto dev $m=16$	2560	12800	0.0082
sae g160softmax	5120	0	0.00436
onto soft ($m=32$)	5120	0	0.00538

Table 1: Reconstruction vs. code capacity, shared eval tail. The plain-SAE frontier dominates the Ontologizer’s deviation code at every measured capacity; the gap is the price paid for contrast-set structure.

5 Results

5.1 The price of structure: reconstruction–capacity Pareto

Findings. (1) At matched FVU_w ≈ 0.49 , the SAE transmits 32 coefficients where the Ontologizer needs 320 (a $10\times$ coefficient-efficiency gap), with the caveat that top- m points are post-hoc truncations of a model never trained to truncate. (2) The Ontologizer’s granularity floor (one deviation per head, 160 coefficients, FVU_w 0.677) is worse than the SAE’s 32-coefficient point — the very sparse regime is structurally closed to it. (3) The parameter- and coefficient-matched dense control (k5120) reaches FVU_w 1.1×10^{-4} with $3.4\times$ fewer active coefficients than the soft Ontologizer: unstructured reconstruction is $\sim 50\times$ better at matched budgets. (4) Head-level codes (mean-ablate all but the m lowest-entropy heads) are far weaker than deviation codes at matched coefficients (heads 8/32: 0.50 vs. dev $m=8$: 0.094): reconstruction information is spread near-evenly across all 160 heads — under the classifier reading, every head’s assignment carries signal for every input, which is what a MoEfied code should look like, and why per-sample head *confidence* does not localize reconstruction. (5) Uniform-origin truncation is strictly worse than $\mathbb{E}[p]$ -origin (dev $m=1$: 0.98 vs. 0.68): the measured mean classification, not uniform, is the code’s true origin; neutral ablations and steering baselines should pin to it.

5.2 Structure decomposition (the ladder)

The grouped-simplex code constraint alone costs $\sim 30\times$ vs. the unstructured dense control (0.0044 vs. 1.1×10^{-4}); everything the Ontologizer adds beyond rung 2b — bilinear classifiers, five residual layers, the shared 2048-d dictionary space — adds *nothing* to reconstruction (slightly negative), at $2.2\times$ the parameters. Hard-per-group selection costs only $\sim 6\%$ over an unstructured code of equal sparsity (g160top1 0.274 vs. k160 0.258), which also shows the Ontologizer’s catastrophic argmax code (FVU_w 17.3; one-hot selection inflates output norm $3.7\times$ because winner entries carry $\sim 4\times$ the soft-mixture norm) is a training artifact, not intrinsic

rung	FVU _w	L ₀	note
k32 baseline	0.499	32	0 dead
g160top1	0.274	160	1434/5120 dead (28%)
g160softmax	0.0044	5120	beats full Ontologizer (0.0054)
k32_p5	0.509	32	deepsup tax ~2%
k32_bl	0.602	32	1848/5120 dead (36%)

Table 2: Converged structural rungs, $m=5120$, 144.3k steps each.

to discrete head codes. Nested prefix losses are nearly free. The bilinear encoder costs $\sim 20\%$ FVU_w and kills a third of the dictionary. Under the steering frame, the ladder’s implication is pointed: *the classifier-forming ingredient is grouped softmax itself*; whether the full stack earns its extra cost must be decided by task naturalness and steering quality, not reconstruction.

5.3 Categorical form

At 1M rows with split-half scoring and 20 size-matched nulls: **g160top1**’s trained partition is *perfectly* head-like (exhaustiveness 1.0000 vs. null 0.64; exclusivity 1.0000 vs. 1.51; sum-CV 0.25 vs. 1.03). Discovery on every unstructured run (**m11264_k32**, **m5120_k32**, **k32_p5**, **m11264_k160**) finds the *opposite* signature — exhaustiveness below null (z from -30 to -50), exclusivity above (z $+13$ to $+31$): spontaneous SAE group structure consists of co-firing topical clusters, not categorical slots, even when the affinity explicitly selects for anti-correlated same-context pairs. And **g160softmax** never fires distinctly at all: no coefficient ever exceeds $2/32$ on any test row — soft grouped codes converge to dense near-uniform mixtures, exactly as the Ontologizer’s soft code does ($\mathbb{E}[p]$ sits 0.06–0.16 bits from uniform in healthy layers; matching soft reconstruction requires ~ 16 of 32 deviations per head). **Categorical structure exists precisely when trained in as hard selection, never emerges from softmax competition under a reconstruction objective, and costs $\sim 60\times$ in FVU_w (0.274 vs. 0.0044).** For steering this means the *soft* architectures’ classifier reading must be cashed out in deviations around $\mathbb{E}[p]$, not in discrete winners.

5.4 Semantic coherence: heads are contrast sets

grouping	groups	mean $z_{\cos}$	frac $z>2$	frac $z<-2$
g160top1 (trained, hard)	160	+161.3	1.00	0.00
k32 discovered (m11264)	359	+1.7	0.36	0.02
onto heads (trained, soft)	160	−17.6	0.04	0.96
g160softmax (trained, soft)	160	−14.7	0.00	1.00

Table 3: Within-head decode-direction coherence vs. size-matched same-pool nulls (**headcoh.py**, $n_{\text{null}}=500$).

This is the sharpest dissociation in the battery, and the one the steering goal predicts. Hard-trained groups are hugely *coherent* (every **g160top1** group far above null: winner-take-all carves the data so each group’s entries tile one decoder cluster — fine-grained substitutions within a region). Post-hoc co-fire clusters are mildly coherent (topical). But both soft classifier architectures are *anti-coherent*: in 96–100% of heads, the entries’ decode directions are

more mutually dissimilar than random same-pool sets. Softmax competition pushes a head’s class decoders apart — within-head entries are well-separated alternatives, i.e. heads function as *contrastive classifiers*, and within-head swaps are large, clean moves along the head’s own dimension. Notably the signature is shared by the Ontologizer and single-layer **g160softmax** with near-identical strength. The choice of inner product is not doing the work: recomputed under the whitened metric the training objective actually optimizes, the repulsion *strengthens* for the Ontologizer (mean z_{cos} -24.4 vs. -17.6 raw; frac $z < -2$ unchanged at 0.96) and is essentially unchanged for **g160softmax** (-14.7) and **g160top1** ($+179.3$ vs. $+161.3$). Two Ontologizer outlier heads (layer-2 h25, $z+346$; layer-3 h28, $z+280$; the next candidates sit in the partially frozen layer 4) cluster instead of contrasting — their sampled entries also decode to generic text, so we suspect partial head collapse (entries never differentiated), not semantic coherence; targeted inspection pending. The description-text view was uninformative on plain summaries ($\sim 77\%$ generic corpus descriptions for tags, 49% for k32 latents, inflating the permutation null: pooled $z = -0.25$); re-run on the contrastive **cacts** descriptions it acquires signal. Within-head description similarity clears the head-label permutation null for **g160top1** ($z=+2.42$, 62 within-head pairs), is marginal for Ontologizer heads ($z=+1.92$, 155 pairs), and is absent for **g160softmax** ($z=+0.79$, 217 pairs). Read against the geometry this is the pattern “alternative values of one shared dimension” predicts: Ontologizer entries’ decode *directions* repel while their described *contents* weakly cohere. It is also the first instrument in the battery to separate the Ontologizer from its single-layer soft rival, whose head contents show no description coherence at all — suggestive rather than significant at this coverage (one global z per model over the campaign’s sampled features), but pointing where the task-reconstruction protocol should look.

5.5 Feature reliability

Seed stability (**k32** vs. **k32_s43**; identical architecture and data, FVU_w 0.4994 vs. 0.4967): matched-feature fractions are **0.31 / 0.24 / 0.15 / 0.03** at $\tau = 0.3/0.5/0.7/0.9$. Roughly three quarters of individual features are not reproducible across initializations at moderate containment. Cross-width splitting (**k32** $\rightarrow$ **m11264_k32**): 190 of 5114 live parents split at $\tau=0.7$ (median 3 children; union coverage 0.37; child overlap 0.42), with 441 absorption candidates and a heavy nearest-neighbour decoder-cosine tail (p50 0.234 vs. random 0.119): splitting is real but is not the dominant organization; topical co-firing is. *Frame note: for a steering generator the unit that must reproduce across runs is the head’s task (partition), not the individual feature vector; partition-level stability is untested and is the right follow-up.*

5.6 Text-space fidelity (collateral ceiling)

The FVU –text mapping is sharply nonlinear: $\text{FVU}_w \approx 0.5$ decodes to same-topic-different-facts text barely above the generic floor, while the Ontologizer’s soft code and **g160softmax** are near the likelihood ceiling — almost all decodable content survives. This matters doubly under the steering frame: a steering primitive that routes through a code which cannot preserve text (the entire $k \leq 160$ sparse tier) starts from ~ 0.75 collateral before any intervention, whereas the dense-mixture classifier codes start near zero. Low-collateral steering is only *available* to the soft architectures.

model	chrF	exact	NLL (ceil. ≈ 1.56 – 1.63 , floor ≈ 3.8)
corpus-mean floor	≈ 0.20	0	3.60–3.89
k32 tier (incl. p5, m11264)	0.23	0.00	3.00–3.08
k32_bl	0.21	0.00	3.19
k160 / g160top1	0.28–0.29	0.00	2.34–2.37
onto soft	0.538	0.090	1.658 (gap 0.026)
g160softmax	0.592	0.119	1.563 (gap 0.007)
k5120	0.856	0.592	1.587 (gap 0.000)

Table 4: Downstream text fidelity, 512 tail rows, forced English decodes.

model	F1@1	F1@4	F1@16	dense
k160	0.119	0.393	0.473	0.676
k32 ($m=5120$)	0.089	0.347	0.465	0.676
m11264_k32	0.078	0.364	0.471	0.676
k32_bl	0.084	0.228	0.368	0.676
g160softmax	0.068	0.162	0.328	0.676
g160top1	0.066	0.200	0.302	0.676
k32_p5	0.061	0.322	0.454	0.676
k5120	0.064	0.130	0.237	0.676
onto (tags)	0.022	0.021	0.117	0.677

Table 5: Macro one-vs-rest F1 over 86 languages, held-out tail.

5.7 Concept (language) localization — off-frame but reported

Language remains distributed in every model (best single-latent macro-F1 0.119 vs. dense ceiling 0.676; script-distinctive languages such as Chinese reach ~ 0.98 from one latent, same-script families smear), and localization *anticorrelates* with reconstruction across the family. The Ontologizer’s tag probabilities are decisively the worst localizers (F1@1 = 0.022). Read on SAE terms this is an indictment; read on classifier terms it is expected: a head encodes a factor in its *assignment*, not in any single tag’s magnitude, and prior partition-level analysis found exactly that — layer-1 heads whose partitions carry script/family mutual information, one of which ((L1,h8,k17)) was causally verified to force Burmese output. Sparse probing and partition MI answer different questions; only the latter matches the architecture. The honest negative that survives the reframe: language factors are carried by a *few* heads’ partitions, not by most, so most heads classify something else — identifying what is the task-reconstruction agenda below.

5.8 Shrinkage decomposition

Top- k SAEs have small magnitude error (share of raw FVU_w removed by support-frozen least squares: 2.2–2.4% for k32 variants, 8.5% for k160, 0.6% bilinear, 1.9% p5); **g160top1** is essentially perfectly calibrated (0.3%) — its reconstruction deficit is pure selection. The Ontologizer’s truncated deviation codes are magnitude-heavy: 16% at $m=1$, 33% at $m=2$, i.e. the pin-and-renormalize convention mis-scales what it keeps (supports of $\geq d$ coefficients are excluded as underdetermined). Separately, the gap between non-adaptive truncation (freeze soft P , truncate, decode linearly) and in-forward truncation (later layers reclassify against the truncated prefix residual) is large — 0.268 vs. 0.094 at $m=8$: downstream layer compensation is worth

$\sim 3\times$ under truncation. This is the one quantified reconstruction credit for the residual stack, and it is steering-relevant: the stack actively re-routes around code edits, which cuts both ways (robustness to small interventions; potential dilution of large ones).

5.9 Steering fidelity (the primary benchmark)

Six arms at magnitudes 0.05–0.5 relative to the unit input norm, 24 features $\times$ 8 held-out contexts $\times$ 6 magnitudes plus 12 random-direction controls — 1,728 steers per arm. Effect is cycle-consistency activation gain normalized by the feature’s q99 over a 16,384-row reference subsample; collateral is $1 - \text{chrF}$ between base and steered decodes.

Collateral is direction-independent. Excess collateral over the random-direction control is within ± 0.03 at every magnitude for every arm except **g160top1** (+0.051 at 0.05, +0.013 to +0.029 elsewhere). Text damage tracks perturbation *magnitude*, not direction: moving an embedding a fixed distance along a learned feature destroys about as much of the decode as moving it that distance at random. Matched magnitude therefore already matches the intervention budget, and effect at matched magnitude — not an effect/collateral frontier — is the comparison.

arm	0.05	0.10	0.15	0.20	0.30	0.50
g160top1 (hard groups)	0.083	0.094	0.073	0.120	0.120	0.156
k32 (decoder rows)	0.010	0.016	0.047	0.083	0.135	0.234
k32_b1 (eigenfeature)	0.036	0.036	0.057	0.094	0.115	0.135
onto resid_nc	0.016	0.021	0.052	0.068	0.078	0.141
onto resid_nc_hm	0.021	0.021	0.021	0.068	0.057	0.083
g160softmax	0.000	0.000	0.000	0.000	0.000	0.010

Table 6: Fraction of steers producing a *strong* effect (> 0.5 of the feature’s q99 activation), by intervention magnitude.

Two regimes, which mean effect conflates. Sparse codes have median effect *exactly* zero at every magnitude — most interventions do nothing, because top- k /hard-group selection returns exact zeros when the feature does not fire — but land hard when they land (mean effect among nonzero steers at $m=0.5$: **k32** 1.63, **g160top1** 1.25, **k32_b1** 0.63). Dense codes move nearly every activation slightly and almost never decisively (**g160softmax** fires “any effect” on 0.88 of steers at $m=0.5$, onto 0.73–0.77, vs. 0.18–0.25 for the sparse arms; but mean effect among those is only 0.19 and 0.26–0.29). Mean effect alone therefore reverses the ranking depending on magnitude and should not be reported by itself.

The grouped-softmax null loses on steering. **g160softmax** — which matches or beats the Ontologizer on reconstruction, code structure, contrastive signature, and text fidelity — essentially *never* produces a strong steer: 0.000 at every magnitude but the largest, where it reaches 0.010. This is the first instrument in the battery on which the full architecture beats its single-layer analogue, and it is the instrument the steering thesis is staked on.

Hard categorical structure is what steers at low magnitude. `g160top1` dominates below $m=0.2$ — $4\times$ the Ontologizer and $8\times$ `k32` at $m=0.05$ — and is the same rung whose trained partition is perfectly head-like (exhaustiveness 1.0000). Categorical form and low-magnitude steerability arrive together, and neither survives softening the selection. It is also the only arm paying collateral above its random control, so the low-magnitude gain is not free. `k32` has the opposite profile: nearly inert below 0.15, best of all arms by 0.5.

Eigenfeature steering has the wrong sign at low magnitude. `k32_b1`’s mean effect is *negative* at $m \leq 0.15$ ($-0.013, -0.015, -0.007$): the closed-form input direction moves the target activation away from itself. This is the causal-side counterpart of the auto-interp result that parameter-space decodes fail as descriptions — a feature’s weight direction is not the direction that excites it.

Caveats. Effect is normalized per feature by `q99`, and dense codes have compressed activation range, so the normalization interacts with code density in both directions; within-regime comparisons (`g160top1/k32/k32_b1`, or the onto arms vs. `g160softmax`) are clean, cross-regime ones need that qualification. Supervised diff-of-means vectors — the natural upper-bound reference — are not implemented in `steerfid.py` and remain unrun. Per-head effect consistency is computable post hoc from `steers.jsonl` but is not reported here.

5.10 Description specificity (auto-interp)

Under the identical plain summarization prompt, 51% of `k32` latents receive a specific description (20% naming a language or script) versus 23% of Ontologizer tags (8%): individual SAE latents are about twice as describable as individual tags, consistent with tags not being standalone features. Blinded detection over the full campaign agrees, and sharpens the reading:

model	params	eig	acts	cacts	<i>n</i>
onto (resid_nc)	0.020	—	0.473	0.392	225
k32	0.158	—	0.645	0.603	250
k32_b1	0.061	0.021	0.578	0.470	75
g160top1	0.039	—	0.646	0.594	150
g160softmax	0.025	—	0.449	0.364	256
k5120 (dense ctrl)	0.001	—	0.470	0.332	256

Table 7: Mean per-feature detection F1 by description mode (always-match null ≈ 0.67 ; a fully generic description that matches every item scores the null, so generic descriptions *inflate* toward it). `k32_b1`’s pool is only 75 live features.

Three readings. (1) *Per-latent describability tracks code density, not architecture.* Activation-based descriptions rank `k32` $\approx$ `g160top1` (0.65) > `k32_b1` (0.58) > onto $\approx$ `k5120` (0.47) > `g160softmax` (0.45): the two hard- $k=32$ codes tie at the top whether or not they carry trained head structure, and the Ontologizer’s tags are exactly as (un)detectable as latents of the near-dense reconstruction-optimal control — per-latent description quality is a property of how many things fire together, not of the classifier structure. This is the detection-side confirmation of the per-latent-lens conclusion. (2) *The contrastive mode is the honest number.* `cacts` scores below `acts` everywhere, but the drop is smallest for the sparse rungs (-0.04 to -0.05) and largest

for the dense control (-0.14): since generic descriptions drift toward the 0.67 null, part of every plain-summary score is generic-match inflation, and the contrastive prompt removes most of it. Under **cacts** the dense codes separate — onto (0.39) edges **g160softmax** (0.36) and **k5120** (0.33) — but the tier ordering stands. (3) *Parameter-space decodes are not descriptions*. M2M100 decodes of parameter directions almost never let the judge detect the feature’s activating texts: **params** is ≤ 0.06 everywhere except plain **k32** (0.158), and the bilinear eigenfeature decode fails identically (**eig** 0.021). Whatever a feature direction decodes to in text space, it is not what fires the feature; naming requires activation evidence. This also lowers the prior on the weight-reading follow-up (per-head eigenfeature decodes) working without an activation-grounded validation step.

5.11 Loss-term ablation: which term rescues layer 4?

The reference run’s frozen layer 4 was attributed to hard-phase head starvation, and **resid_nc_hm** added two loss terms at once to fix it — a head-similarity penalty ($s_{h\text{cossim}}=10^{-5}$) and a batch mean-entropy bonus ($s_{Hm}=10^{-6}$, $\text{KL}(\mathbb{E}_{\text{batch}}[p] \parallel \text{unif})$). Neither could be credited. We train the missing cell, **resid_nc_hc** ($s_{h\text{cossim}}$ only), from an initialization bit-identical to **resid_nc_hm**’s, and compare all three at the matched step 370,000 (one 24-epoch pass is 372,600 steps).

Evaluation adds a *fresh-encoded* out-of-distribution set, which also discharges the “fresh-encoded evaluation set” caveat of the Setup: 780,889 messages from an AI-and-human community Discord export ([**TODO: name/cite the corpus**]; 74.8% bot-authored, 2024-07–2025-04) encoded through the same frozen SONAR encoder; deduplicated by exact message text (13.5% dropped) and scored against a duplicate-free mC4 tail window. Neither set was trained on by any arm.

run	layer 4 code health			layer 4 FVU _w		
	eff. ent.	collapsed	dead	mC4	fresh	ratio
resid_nc (neither)	3.96	28/32	857	0.00532	0.05423	10.2×
resid_nc_hc ($s_{h\text{cossim}}$)	19.27	0/32	0	0.00183	0.04851	26.6×
resid_nc_hm (both)	31.06	0/32	0	0.00190	0.04927	25.9×

Table 8: Loss-term decomposition at matched step 370,000. Effective entries = $\exp H(\mathbb{E}[p])$ per head (max 32); collapsed = heads with < 10 ; dead = entries of 1024 with marginal mass $< 10^{-4}$. “fresh” is the deduplicated Discord set.

$s_{h\text{cossim}}$ **does all the rescuing and all the damage**. It alone eliminates the freeze — 28/32 collapsed heads and 857 dead entries go to zero — and alone accounts for the entire in-distribution gain (final whitened MSE $4.42 \rightarrow 3.53 \times 10^{-5}$; adding s_{Hm} gives 3.56×10^{-5} , no better) and the entire head decorrelation (cossim_h $3.87 \rightarrow 0.83$; with s_{Hm} , 0.90). It is also solely responsible for the deep-layer generalization gap: the OOD ratio climbs identically with or without s_{Hm} (26.6× vs. 25.9×), whereas **resid_nc**’s apparently flat profile (3.7/6.8/11.3/11.0/10.2 across prefixes) is an artifact of a dead layer that cannot diverge because it barely contributes — its layer-4 FVU_w is *worse* than layer 3’s.

s_{Hm} **buys interpretability, not reconstruction**. What it prevents is milder and different: progressive narrowing of entry usage with depth. Without it, effective entries fall monotonically

(31.5, 30.8, 29.5, 26.6, 19.3 across layers); with it they stay flat (31.6, 31.1, 30.8, 30.9, 31.1). This is what the rising KL_m trace measures — not incipient freezing but an increasingly lopsided usage distribution. The downstream effect is confined to the properties the steering frame cares about: at layer 4, s_{Hm} raises entry usage 61%, mutual information with message length $0.058 \rightarrow 0.101$ nats, and length-controlled within-entry semantic coherence $+0.0046 \rightarrow +0.0077$ — at no reconstruction cost and no transfer cost. If only reconstruction matters, s_{Hm} is unnecessary; it is the term that keeps the deep layers legible.

Transfer is not overfitting. Across training, held-out-corpus FVU_w on the fresh set falls monotonically for `resid_nc_hm` — 3.55, 0.180, 0.080, 0.057, 0.052, 0.039, 0.040, 0.029 at steps 10k, 50k, 100k, 300k, 370k, 1M, 2M, 3.05M (unduplicated, 10,240 rows, so not directly comparable with the table) — and never rises. The widening ratio is a shrinking denominator, and the mC4 trace is itself non-monotone (0.0019 at 370k, 0.0136 at 2M, 0.00047 at 3.05M), so the ratio should be read alongside absolute numbers. Depth buys refinement that is increasingly corpus-specific but never harmful.

6 Formal reading: heads as causal variables

The steering frame has a natural formalization in the causal-abstraction vocabulary, and stating it makes several instruments in the battery estimators of named quantities rather than ad hoc scores.

Definitions. Read the Ontologizer as a structural causal model: each head (l, h) is a candidate high-level *variable* with value space $\{1, \dots, k\}$ (soft state: its simplex), the residual stack is the causal graph (with residual forwarding, layer $i+1$ ’s classifier reads the residual written by layers $\leq i$, so every earlier head is a parent of every later one), and `withArgs` set-interventions are the do-operator: $\text{do}[(l, h)=k']$ forces the head’s assignment while leaving all non-descendants untouched by construction. The abstraction map τ sends an embedding x to its per-head assignment profile $P(x)$. Under this reading: steerfid’s cycle-consistency gain is an estimator of *interchange-intervention accuracy* in the closed loop $\tau \circ \text{encode} \circ \text{decode}$ — did forcing the variable produce a decode that re-encodes to the forced value? — and `splitting.py`’s seed-to-seed match fractions are *empirical identifiability*: how much of the learned variable set is determined by the data rather than the seed. What the softmax leaves unidentified in principle is the per-head entry order (permutation) and any logit gauge the temperature absorbs; what must therefore be reproducible for the variable reading to hold is the partition itself and the decode directions up to relabeling.

Metric hygiene. Claims about decode-direction geometry should not depend on an arbitrary choice of inner product. The contrast-set result does not: under the training objective’s inverse-variance metric the Ontologizer’s within-head repulsion strengthens (-24.4 vs. -17.6 raw) and both rivals’ signatures are unchanged (details in the semantic-coherence subsection; artifacts in the `headcoh_w` companion runs).

Composition. A variable reading earns little if interventions on two heads interact uncontrollably. Measured additivity of decode deltas (`compose.py`, whitened metric, medians over pairs):

stratum	n	cos	rel. err	note
same-layer L0	28	0.959	0.350	4 downstream layers
same-layer L1	28	0.947	0.354	3
same-layer L2	28	0.967	0.259	2
same-layer L3	28	0.996	0.092	1
same-layer L4	28	1.000	0.019	0 (= numerical floor)
cross-layer (earlier = L0/L1)	42	0.97–0.98	0.20–0.23	
cross-layer (earlier = L2/L3)	18	1.00	0.02–0.04	

Three observations. (i) Joint interventions point where the summed singles point everywhere (median cosine ≥ 0.95 in every stratum): heads compose approximately additively, which is what the steering-primitive claim needs. (ii) The non-additive component is entirely attributable to downstream reclassification: it decays monotonically with the number of layers below the intervention and hits the numerical floor when there are none — the final-layer stratum is provably exact (the decode is linear in the flattened classification), and the measured 0.019 there is float32 rounding amplified by the $\|R\| \approx 31 \rightarrow \|Y\| \approx 1$ decoder cancellation, verified $\sim 3 \times 10^{-6}$ in float64. (iii) Cross-layer pairs compose *better* than same-layer pairs at matched depth: a set-intervention is immune to upstream perturbation (it forces the head regardless of input), so cross-layer interaction is one-way, while two same-layer interventions’ perturbations compound through the same downstream stack. For steering practice this means multi-head edits are cleanest either in one late layer or spread across layers — and interaction strength, 1–additivity, is itself a measure of how far the residual stack is from a bag of independent classifiers, i.e. of how much the hierarchy follow-up (conditional MI across layers) has to explain.

7 Preliminary conclusions

1. **Contrast-set structure is real, and it is what the architecture buys.** Trained softmax heads carry the exact signature the steering goal predicts: within-head decode directions repel (96% of heads below null), so each head holds well-separated alternatives — contrastive classes — rather than a semantic cluster. This is invisible to reconstruction metrics and to per-latent probes, and it was the design’s central bet.
2. **Reconstruction is the price sheet, not the verdict.** Unstructured codes reconstruct $\sim 50\times$ better at matched budgets and the grouped-simplex constraint alone costs $\sim 30\times$. The architecture cannot be motivated by reconstruction; it must be motivated by what the structure enables — nameable tasks and effective low-magnitude steering. The steering half is now measured and comes out in the architecture’s favor; task naturalness is still unrun.
3. **The strongest rival is inside the ladder.** Single-layer `g160softmax` reproduces the code structure, the contrastive signature, and the text fidelity of the full Ontologizer at better reconstruction with half the parameters. Its heads are *not*, however, equally steerable — it is the weakest arm in the steering battery, which is the first instrument to separate the two — so the open question narrows to task naturalness; the stack’s measured credits — $3\times$ adaptive compensation under truncation, and a first suggestive crack in the rival’s favor-sheet: Ontologizer head contents weakly cohere in description space ($z+1.9$)

where `g160softmax`’s do not ($z+0.8$) — and its hierarchical task structure (deeper heads classify over shallower assignments — untested) are what remain to defend it.

4. **Soft classifier codes are dense-mixture codes.** Softmax competition under reconstruction converges to near-uniform mixtures in both architectures; genuine categorical form exists only when selection is hard (then it is textbook-perfect and cheap relative to equal sparsity: $\sim 6\%$). Steering primitives for the soft architectures must therefore operate on deviations around the measured origin $\mathbb{E}[p]$ — which decodes to the corpus mean — and ablations must pin to it, never to uniform. Hard selection buys more than categorical form: `g160top1` is also the best low-magnitude steerer, $4\times$ the Ontologizer and $8\times$ plain top- k at $m=0.05$. Categorical structure and low-magnitude steerability arrive together and neither survives softening.
5. **Per-latent lenses mis-measure classifier codes.** k -sparse probing ranks the Ontologizer worst while partition-level MI finds verified causal language heads; plain per-feature summaries call 77% of tags generic; blinded detection F1 ranks codes by density, with the Ontologizer indistinguishable from the near-dense control (`acts` 0.47 both) despite entirely different structure. Evaluation of grouped codes needs partition-level and contrast-level instruments (headcoh, partition MI, sibling-contrastive descriptions) — a methodological finding in its own right.
6. **Collateral is a magnitude budget, not a direction choice.** Steering along a learned feature destroys about as much of the decode as moving the same distance at random (excess collateral within ± 0.03 for every arm but `g160top1`), so “low-collateral steering” is not a property a better direction can buy — only a smaller magnitude, which makes effect at low magnitude the whole game. This compounds with a floor: the entire sparse tier ($FVU_w \gtrsim 0.25$) decodes to same-topic-different-facts text, i.e. ~ 0.75 collateral before any intervention. The dense-mixture classifier codes (onto, `g160softmax`) sit near the likelihood ceiling. Whatever wins the steering comparison must come from that tier or fix the SAE tier’s fidelity first.
7. **Reliability must be re-based to tasks.** Individual features reproduce poorly across seeds ($\sim 24\%$ at $\tau=0.5$); “natural clusters” predicts that head *partitions* reproduce even when feature vectors do not. That prediction is testable and untested.

8 Follow-up investigations

Ordered by relevance to the steering goal.

- **Task reconstruction (the core agenda).** Recover what each head classifies: (a) partition-vs-label-library MI (language/script from the sidecar, length, formatting/noise, unsupervised topic clusters at several granularities) — quota-free; (b) *sibling-contrastive* descriptions — for entry (h, k) , positives are rows where k wins within head h , background are rows where its *siblings* win, so the judge names the decision content, followed by a head-level pass naming the single question the k classes partition, validated by blinded assignment accuracy vs. $1/k$; (c) classifier weight reading: per-head eigenfeatures of the bilinear classifier decoded through M2M100, with the eigenvalue spectrum as task dimensionality; (d) FLORES parallel-text invariance splitting content heads from form heads

with no LLM. Run identically on Ontologizer heads and `g160softmax` groups: whether the full stack finds *more natural* tasks than one grouped-softmax layer is the live architectural question.

- **The steering overlay — remaining pieces.** The magnitude sweep and random-control curves are run (see Steering fidelity). Still open: supervised diff-of-means vectors as an upper-bound reference (not implemented in `steerfid.py`); per-head effect *consistency* across contexts, computable post hoc from `steers.jsonl`; and a q99-free effect normalization, since the current one interacts with code density. Note the effect-per-collateral framing is moot: collateral is direction-independent at matched magnitude.
- **Partition-level seed stability.** Train a second Ontologizer (or `g160softmax`) seed; match heads across runs by partition MI and report the distribution — the “natural clusters” hypothesis predicts partition agreement despite $\sim 24\%$ feature-level reproducibility. `splitting.py` on two `g160top1` seeds answers the group-content version.
- **Hierarchy as MoE routing.** Layer- $i+1$ heads classify over layer- i assignments: conditional-MI between successive layers’ partitions tests whether depth builds compositional tasks (classifiers of classifications) — the one capability no single-layer variant can replicate in principle, and currently the stack’s main untested defense.
- **Fresh-encoded eval set.** Removes the onto-trained-on-the-tail asymmetry; upgrades every table.
- **Winner dropout for grouped rungs.** `g160top1`’s 28% dead latents are within-group winner-take-all starvation; the Ontologizer’s winner-dropout mechanism is the known fix. Also worth testing: batch mean-entropy pressure (s_{Hm} -style) on group usage.
- **Trained truncatability.** A deviation-sparsity penalty (or matryoshka-over-deviations) at training time would show whether the onto Pareto curve can shift left — and whether sparser contrastive codes steer better.
- **Rung composition and frontier fill.** `groups+prefixes` (2b+3), grouped variants at $m=11264$, the mid-range top- k sweep ($k \in \{512, 2048\}$).
- **Refit fairness.** Random-support controls at matched K (especially $K \gtrsim d/2$), and a nonnegative-LS variant matching each code’s sign constraints.
- **Coherent-head outliers.** Inspect L2h25/L3h28 (the two anomalously coherent heads) for partial collapse. The layer-4 freeze itself is resolved: the head-similarity penalty, not the batch mean-entropy bonus, is the effective fix (see the loss-term ablation). Open: separating $s_{hcossim}$ ’s rescue from its deep-layer transfer cost, e.g. annealing it off after the hard phase.

Reproducibility

All protocols are root-level scripts in the `ontologize` repository (`sae.py`, `pareto.py`, `headstruct.py`, `headcoh.py`, `splitting.py`, `textfid.py`, `langprobe.py`, `refit.py`, `steerfid.py`, `compose.py`,

autointerp.py), driven by the `sae_ladder.sh`, `sae_evals.sh`, and `autointerp_run.sh` runners (idempotent, crash-resumable systemd user units; judge calls rate-metered). The curated test suite (119 tests) pins each protocol’s statistics against planted-structure synthetics. Artifacts live under `data/out/sonar/{pareto,refit,splitting,textfid,langprobe,steerfid,autointerp}` and per-run `headstruct/ / headcoh/ / compose/` directories.

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